

Yaswanth Maddela (Data Engineer)

Phone No: (573) 381-0776 Email ID: yaswanthm1412@gmail.com Location: Illinois, USA



Proactive Data Engineer with 3+ years of experience in designing, developing, and optimizing large-scale data pipelines and architectures. Adept at leveraging big data technologies, cloud platforms, and ETL processes to drive business insights and efficiency.

PROFILE SUMMARY

- Senior Data Engineer with 3+ years of experience designing, developing, and optimizing large-scale data architectures, ETL pipelines, and cloud-based solutions for diverse industries.
- Expert in cloud platforms including AWS, Azure, and GCP, leveraging services such as S3, Redshift, Lambda, Glue, EMR, Data Lake, Data Factory, Synapse Analytics, BigQuery, Dataflow, and Pub/Sub for scalable, cost-efficient data solutions.
- Proficient in modern data engineering tools and frameworks, including Apache Spark, Hadoop, Hive, Kafka, Airflow, NiFi, Talend, Informatica, and AWS Glue, for batch and real-time data processing.
- Strong database and programming expertise with SQL, MySQL, PostgreSQL, MongoDB, Cassandra, Scala, and Python, enabling optimized data modeling, query performance, and automated workflows.
- Advanced experience in CI/CD and DevOps practices using Git, GitHub Copilot, Maven, Jenkins, Docker, and Kubernetes for version control, pipeline automation, and continuous delivery.
- Specialized in AI, ML, NLP, and Generative AI integration within data pipelines, including predictive modeling, text analytics, LLMs, and end-to-end machine learning workflow automation.
- Skilled in building real-time and batch data pipelines for analytics and business intelligence using Kafka, Airflow, Spark Streaming, and AWS Lambda, alongside cloud-native orchestration tools.
- Proficient in data warehousing and visualization, implementing Redshift, Snowflake, BigQuery, and Azure Synapse, alongside visualization tools like Tableau and Power BI to deliver actionable insights.
- Experienced in hybrid and multi-cloud architectures, ensuring secure, reliable, and high-performance data ingestion, processing, and storage across distributed systems.
- Forward-looking data engineering thought leader, leveraging emerging technologies like GitHub Copilot, LLMs, NLP, and Gen AI for innovative, automated, and intelligent data solutions.

RELEVANT SKILLS

- **Cloud Platforms & Services:** AWS, GCP, Azure; AWS S3, Athena, Glue, EMR, Redshift, Data Lake, Lambda, Kinesis; Azure Data Lake, ADF, Databricks, Azure SQL; GCP BigQuery, Dataflow, Pub/Sub, Cloud Storage
- **Databases & Data Warehousing:** Oracle, MySQL, SQL Server, PostgreSQL, HBase, Snowflake, Cassandra, MongoDB; OLAP, Dimensional Modeling, Schema Design, Data Marts, Redshift, Snowflake, BigQuery, Azure Synapse Analytics
- **Programming & Scripting Languages:** Python, Java, Scala, Shell Script (Bash, Shell), Hibernate, JDBC, JSON, HTML, CSS; Python for data pipelines, automation, ML/NLP workflows, and API integrations
- **Big Data & Distributed Systems:** HDFS, Hue, MapReduce, PIG, Hive, HCatalog, HBase, Sqoop, Impala, Zookeeper, Flume, Kafka, YARN, Cloudera Manager, Kerberos, PySpark, Spark Streaming, Airflow
- **ETL & Data Integration Tools:** Informatica, Talend, AWS Glue, Apache NiFi, Airflow, dbt, Data Migration, API Integration, Data Pipelines
- **Visualization & BI Tools:** Tableau, Power BI, Looker, QuickSight
- **DevOps, Version Control & CI/CD Tools:** Git, GitHub Copilot, Maven, SBT, CBT, Jenkins, Docker, Kubernetes, Terraform
- **AI, ML, NLP & Gen AI Skills:** Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), Generative AI, AI-powered Data Pipeline Automation, Predictive Analytics, Model Deployment
- **Operating Systems & Web Servers:** Windows, Linux, Apache Tomcat, WebLogic
- **Methodologies & Collaboration Tools:** Agile (Scrum), Waterfall, UML, Design Patterns, SDLC; JIRA, Confluence, Slack, Microsoft Teams
- **Networking & Cloud Management:** Data Integration, Cloud Networking, API Management, Data Migration, Hybrid & Multi-Cloud Architecture

RELEVANT EXPERIENCE

Azure Data Engineer - Bank of Montreal, Chicago, Illinois, USA

Feb 2025 – Dec 2025

The Bank of Montreal (BMO), is a major Canadian chartered bank with global operations, offering a wide range of financial services. I am developing data processing workflows using tools like Azure Databricks or Azure Synapse Analytics to transform, clean, and aggregate data according to business requirements.

Responsibilities:

- Designed and implemented scalable Azure Data Factory pipelines to automate ingestion, transformation, and orchestration of structured and unstructured banking data for real-time and batch analytics across retail, commercial, and wealth banking systems.
- Leveraged Azure Synapse Analytics, Apache Kafka, and Azure Event Hubs to process transactional and operational data with high concurrency and low latency, enabling actionable insights for risk management, fraud detection, and customer analytics.
- Engineered cloud-native storage solutions using Azure Data Lake, Azure Blob Storage, and Snowflake, optimizing data access and retention for regulatory compliance, risk modeling, and predictive analytics.
- Developed advanced SQL, T-SQL, and PostgreSQL queries and complex transformations to support credit risk scoring, AML/KYC compliance, fraud detection, and customer behavior analytics.
- Implemented Azure Databricks and Apache Spark for large-scale data processing, integrating machine learning models for customer segmentation, predictive loan default analysis, and personalized marketing strategies.
- Automated CI/CD pipelines for data engineering workflows using Azure DevOps, Git, Maven, and GitHub Copilot, ensuring rapid, error-free deployment of cloud data solutions.
- Built interactive dashboards and visualizations with Power BI, Tableau, and Plotly, providing executive stakeholders with real-time KPIs and insights for data-driven decision-making.
- Integrated AI, ML, NLP, LLM, and Generative AI models within data pipelines for predictive analytics, automated anomaly detection, fraud prevention, and customer personalization across banking channels.
- Developed ETL pipelines using Apache NiFi, Talend, and Azure Data Factory, orchestrating data movement across on-premises, cloud, and third-party banking platforms, while leveraging Sqoop, Flume, and Zookeeper for large-scale ingestion.
- Containerized and orchestrated scalable data services using Docker and Kubernetes, managing structured and unstructured datasets with MySQL, MongoDB, Cassandra, and Scala, ensuring high availability, performance, and unified analytics for Bank of Montreal's international banking operations.

AWS Data Engineer - BlackRock, Mumbai, India

Jan 2023 - Nov 2023

BlackRock, Inc. is an American multinational investment company. Involved in designing and building scalable data pipelines using AWS services such as AWS Glue, Amazon Redshift, and S3. Developed efficient ETL processes for data extraction, transformation, and loading into data warehouses and lakes.

Responsibilities:

- Designed and implemented scalable ETL pipelines using AWS Glue, AWS Lambda, Amazon S3, Apache NiFi, and Talend to process, transform, and integrate insurance and financial datasets for underwriting, claims management, and risk analytics.
- Built real-time data ingestion and streaming solutions with Amazon Kinesis, AWS Lambda, and Apache Spark Streaming, enabling fraud detection, risk assessment, and operational efficiency in insurance and asset management workflows.
- Architected and optimized cloud data warehouses using Amazon Redshift, Snowflake, and Amazon Aurora, supporting regulatory reporting, advanced analytics, and large-scale financial performance monitoring.
- Managed structured and unstructured data with Amazon RDS, DynamoDB, Amazon Neptune, and Hadoop ecosystem tools (Sqoop, Impala, Flume, Zookeeper), ensuring secure, scalable, and high-performance access for analytics and reporting.
- Automated end-to-end data workflows and orchestration using AWS Step Functions, AWS Data Pipeline, Apache Airflow, and CI/CD tools (Jenkins, GitLab CI, Docker, Kubernetes), enabling rapid deployment, scaling, and governance of AWS data solutions.

- Integrated third-party and regulatory data sources via AWS API Gateway, Amazon AppFlow, and Lambda, ensuring seamless interoperability, compliance, and standardized data pipelines across financial systems.
- Developed actionable dashboards and reports with Power BI, Tableau, and Python visualizations, providing executive stakeholders with insights on underwriting performance, claims trends, customer analytics, and portfolio risk.

Data Engineer - Astellas Pharma, Mumbai, India

Jun 2021 - Dec 2022

Astellas Pharma Inc. is a Japanese multinational pharmaceutical company. Utilized ETL tools and frameworks to automate data workflows, ensuring timely availability of data for analytics

Responsibilities:

- Designed and implemented scalable ETL pipelines using Apache Spark, Apache Kafka, Apache NiFi, and Talend for real-time and batch processing of structured and unstructured pharmaceutical and clinical data, ensuring efficient ingestion, transformation, and integration across research, regulatory, and operational systems.
- Developed and optimized data storage and querying solutions using Hadoop HDFS, Hive, MongoDB, Cassandra, and NoSQL/SQL databases, enabling fast access, high concurrency, and secure management of large-scale clinical, patient, and operational datasets.
- Built advanced analytics and predictive modeling workflows using Python, PySpark, Scikit-learn, and TensorFlow, applying machine learning, NLP, and AI/ML techniques for risk assessment, clinical insights, and operational efficiency, while enhancing data-driven decision-making.
- Automated, orchestrated, and deployed data pipelines using Apache Airflow, Docker, Kubernetes, and Jenkins, ensuring scalable, fault-tolerant, and CI/CD-enabled workflows for pharmaceutical data applications, including clinical trials and regulatory reporting.
- Implemented data governance, profiling, and compliance frameworks using Apache Atlas and Collibra, ensuring adherence to GDPR, HIPAA, and other regulatory standards.

EDUCATION

Southern Illinois University - Illinois, USA

- Masters in Computer Science (Jan 2024 - Dec 2025)